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THE MCWHIRTER CRITERION REVISITED FOR LASER-INDUCED PLASMAS

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CONTEXT and OBJECTIVES

The McWhirter criterion

When electron-induced collisions are dominant in a plasma, the McWhirter criterion is used to estimate the minimum electron density n_e required for Local Thermodynamic Equilibrium (LTE). LTE is satisfied if

$$n_e > A\sqrt{k_B\, T_e}\,(\Delta\varepsilon_{ki})^3$$

with $A = 1.6 \times 10^{18}\, m^{-3} eV^{-7/2}$, k_B is the Boltzmann constant, T_e is the electron temperature and $\Delta\varepsilon_{ki}$ is the maximum value of the energy difference between the adjacent levels k and i connected by an optically allowed transition [1].

Questions

- What role does radiation play in departure from LTE?
- Can this criterion be refined?

} for a **laser-induced plasma on Aluminum**

COLLISIONAL–RADIATIVE MODEL

ECHREM code [2]

- *Species*..... Al, Al⁺, Al²⁺, electrons
- *Excited electronic states*... **Al** → 64 states $\Delta\varepsilon_{ki} = 3.13\ eV$, $\varepsilon_{ioni} = 5.986\ eV$
  Al⁺ → 124 states $\Delta\varepsilon_{ki} = 4.66\ eV$, $\varepsilon_{ioni} = 18.829\ eV$
  Al²⁺ → 1 state
- *Elementary processes*
  - Electron-induced excitation-deexcitation [2]
  - Electron-induced ionization-recombination [2]

  } detailed balance implemented
  - Spontaneous emission [3] with escape factors (Stark and Doppler broadenings) [4]

⇒ Time-dependent kinetic equation solved for the density $[X_k]$ (species X on the state k)
- Hemispherical plasma
- Radius R

TEST-CASE no 1

Relaxation

- Recombination under almost constant n_e
- Constant temperature $T_e = 10\,000\ K$ (order of magnitude for **laser-induced Aluminum plasma**)
- Constant radius $R = 1\ mm$ (order of magnitude for **laser-induced Aluminum plasma**)
- BOLTZMANN and SAHA-BOLTZMANN plots for $t = 1\ \mu s$

- According to McWhirter:
  - For Al → $n_e > 4.6 \times 10^{19}\ m^{-3}$
  - For Al⁺ → $n_e > 1.5 \times 10^{20}\ m^{-3}$

  } ⇒ LTE

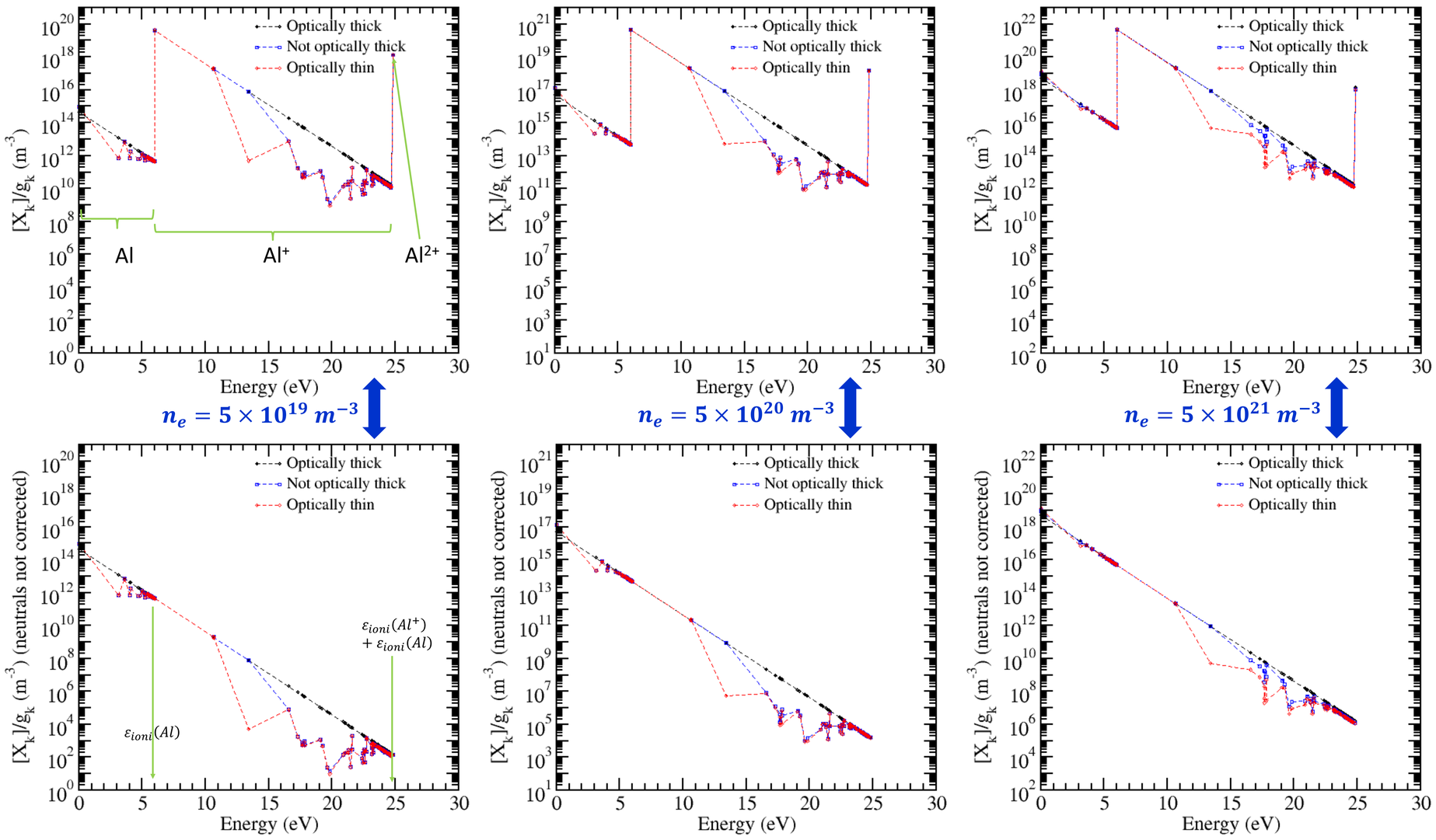
⇒ **3 calculations** performed at $5 \times 10^{19}, 5 \times 10^{20}, 5 \times 10^{21}\ m^{-3}$

BOLTZMANN plots

The excitation equilibrium can be verified

SAHA-BOLTZMANN plots

*The excitation **and** ionization equilibrium can be verified*



$n_e = 5 \times 10^{19}\ m^{-3}$
- Al → **not in LTE**
- Al⁺ → **not in LTE**

$n_e = 5 \times 10^{20}\ m^{-3}$
- Al → **in LTE**
- Al⁺ → **not in LTE**

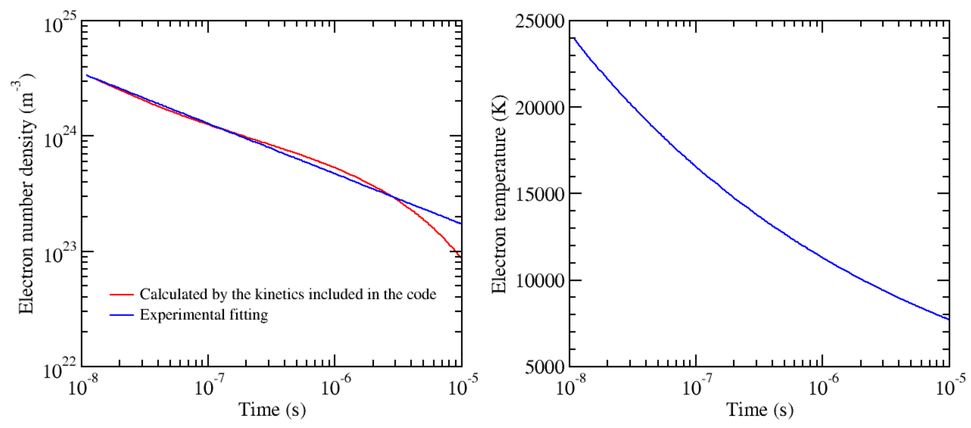
$n_e = 5 \times 10^{21}\ m^{-3}$
- Al → **in LTE**
- Al⁺ → **not in LTE**

For Al and Al⁺, the McWhirter criterion strongly underestimates the lowest electron density to reach LTE by one order of magnitude at least.

TEST-CASE no 2

Experimental time-evolution (typical conditions for a **laser-induced Aluminum plasma**)

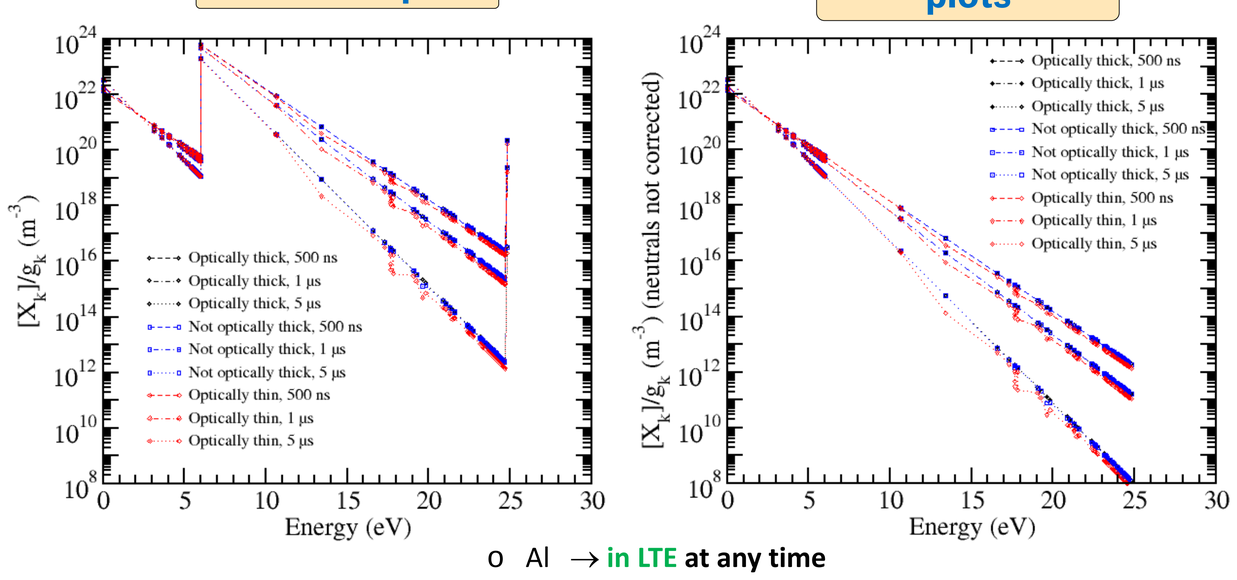
- Electron density $n_e(t) = 3.5 \times 10^{24} \left(\frac{t[ns]}{10}\right)^{-0.436}\ m^{-3}$ for $t > 10\ ns$
- Electron temperature $T_e(t) = 24\,270 \left(\frac{t[ns]}{10}\right)^{-0.166}\ K$ for $t > 10\ ns$
- Plasma radius $R(t) = 0.63 \left(\frac{t[ns]}{10}\right)^{0.1}\ mm$ for $t > 10\ ns$



The code calculates $n_e(t)$ which correlates very well with experiment

BOLTZMANN plots

SAHA-BOLTZMANN plots



- Al → **in LTE at any time**
- Al⁺ → **very close to LTE**

In typical laser-induced Aluminum plasma conditions, the electron density strongly exceeds the McWhirter criterion: LTE is fulfilled.

CONCLUSIONS & PERSPECTIVES

- **Detailed view of the dynamics of an Aluminum plasma in recombination through CR modeling**
- **For Al and Al⁺, the McWhirter criterion strongly underestimates the n_e threshold to reach the LTE**
- **Even at high n_e value, radiation plays a role in the excited states distribution**
- **A new criterion will be identified including the role of (1) radiation and (2) expansion**

REFERENCES

[1] R.W.P. McWhirter, Plasma diagnostic techniques, Academic Press, 1965
[2] V. Morel, A. Bultel, et al., Spectrochimica Acta Part B 103-104 (2015) 112-123
[3] NIST Database
[4] T. Holstein, Physical Review 72, 12 (1947) 1212-1233